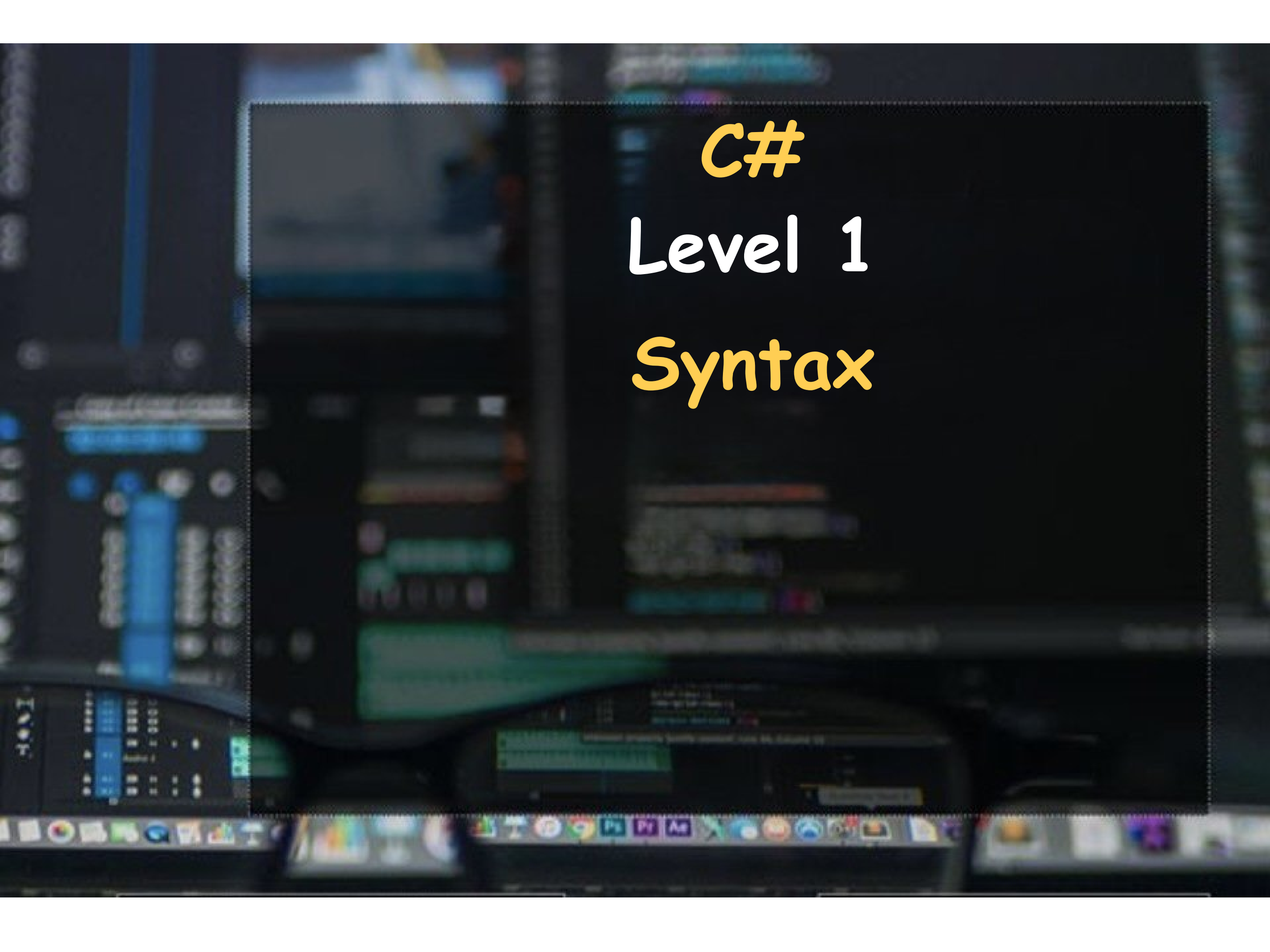
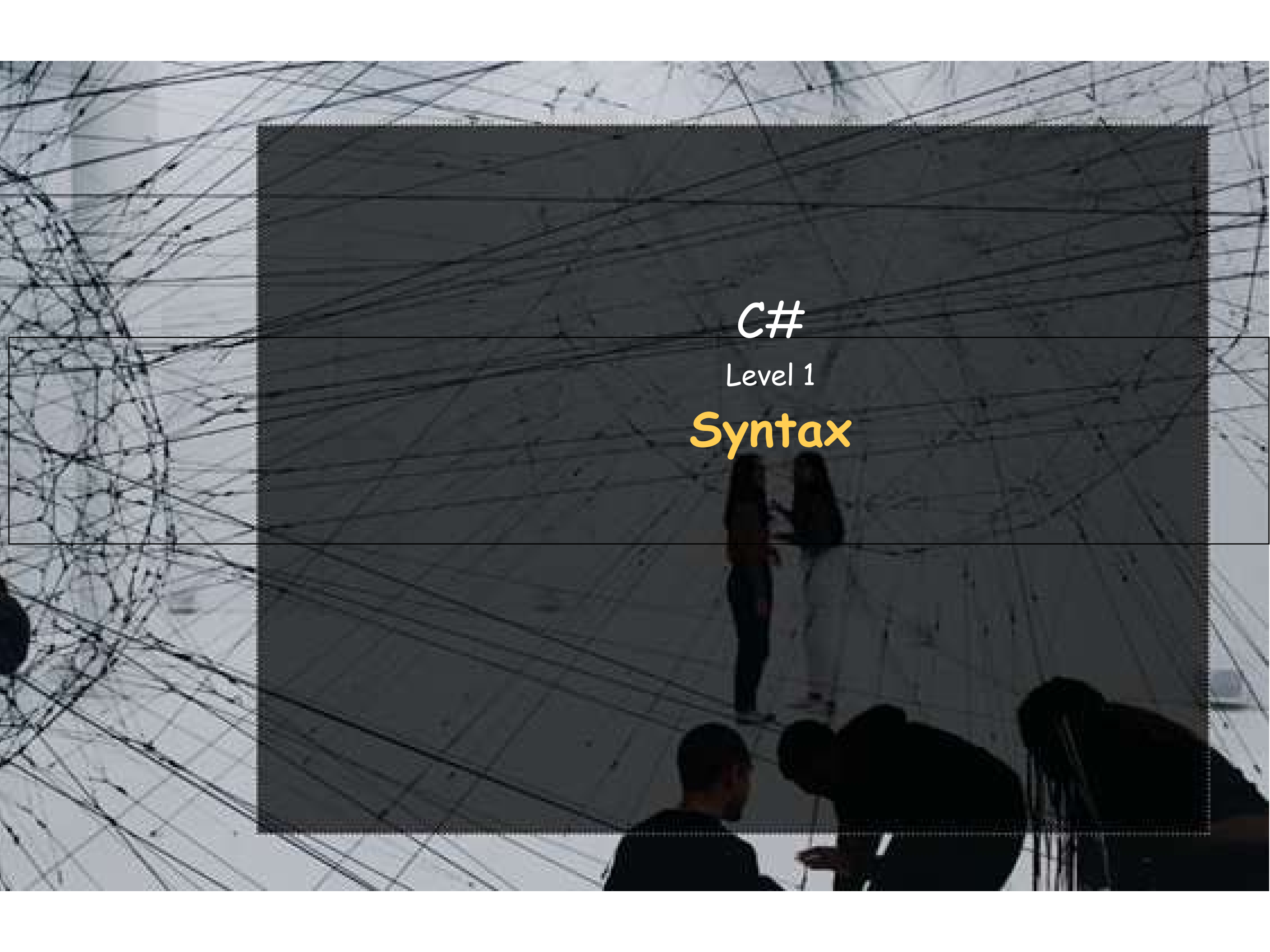




C# Level 1 Syntax



C#

Level 1

Syntax

Definition

C# Syntax

 Syntax is the set of rules that defines how *C#* code must be written.

 It includes elements such as using directives, namespaces, classes, methods, and statements.

Explanation

Using Directive

-  ``using System;`` allows types such as ``Console`` to be referenced without writing the full namespace.

Namespace

-  Organizes related classes and types.
-  Helps avoid naming conflicts.

Class

-  Defines a type that can contain data and methods.
-  ``internal`` means accessible within the same assembly.

Main Method

-  ``Main()`` is the traditional entry point of a console application.
-  ``static`` means it belongs to the class.
-  ``void`` means it returns no value.
-  ``string[] args`` receives command-line arguments.

Console Methods

-  ``Console.WriteLine()`` displays output.
-  ``Console.ReadLine()`` reads a line of text.

Procedures

1. Add required `using` directives.
2. Define a namespace if needed.
3. Create a class.
4. Define the `Main()` method.
5. Write program statements.
6. Compile and execute the program.

💡 Example

```
using System;
namespace ConsoleApp1
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("Hello World!");
            Console.ReadLine();
        }
    }
}
```

➡ Result
Hello World!

Note

- ◆ `{}` define code blocks.
- ◆ `()` are used with methods and parameters.
- ◆ `;` normally terminates a statement.
- ◆ *C#* is case-sensitive.

⊘ Warning

- ◆ ``Console`` and ``console`` are different.
- ◆ Missing ``{}`` can cause structural errors.
- ◆ Missing ``;`` can cause compilation errors.
- ◆ Modern *C#* also supports top-level statements, so ``Main()`` is not always written explicitly.



Comparison



`Console.WriteLine()`

- ✓ Displays output.
- ✓ Moves to a new line.



`Console.ReadLine()`

- ✓ Reads user input.
- ✓ Returns the input as a string.

✦✦ Summary

- ✓ Syntax defines the structure of *C#* code.
- ✓ ``using`` provides easier access to namespaces.
- ✓ ``namespace`` organizes types.
- ✓ ``class`` defines a type.
- ✓ ``Main()`` is the traditional entry point.
- ✓ ``WriteLine()`` displays output.
- ✓ ``ReadLine()`` reads input.

Thank you 😊